CCCA Cell-State Analysis - Zhang et al. 2019 - Follicular Lymphoma

GO / NO-GO GATES

Gate 1 - sample mixing & manifold continuity — GO

All 4 samples (2 patients) intermix on one continuous connected manifold at Harmony theta=10 (theta=2 = per-patient archipelago); 0 detached islands; iLISI 1.04 -> 2.26 (max 4).

Gate 2 - embryonic module geography — GO (non-branching)

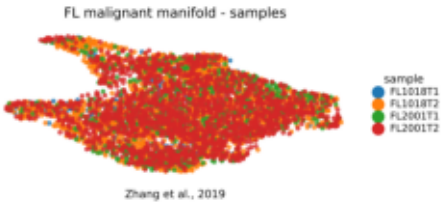
Gast/pSom weakly overlap at one pole (r=-0.16); Crest opposes both (r=-0.54 / -0.57) as an inverse terminal gradient rising along pseudotime (rho=+0.33). No competing arms; no Crest inversion.

Gate 3 - hub Gast<->pSom cross-repression — NA

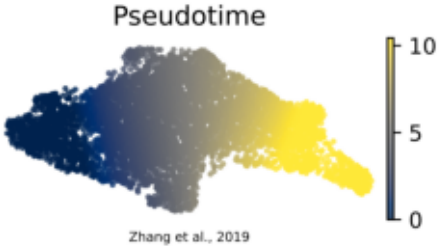
Non-branching manifold - single developmental axis, no bifurcation and no convergence hub, so the branching-only cross-repression test does not apply.

STEP FIGURES

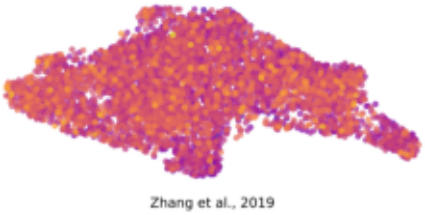
01 Gate 1: sample integration



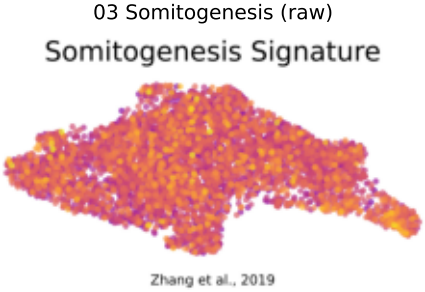
02 Trajectory (pseudotime, curve)



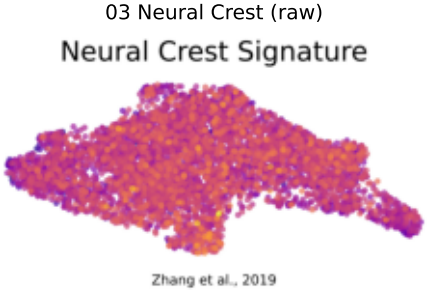
03 Gastrulation (raw)



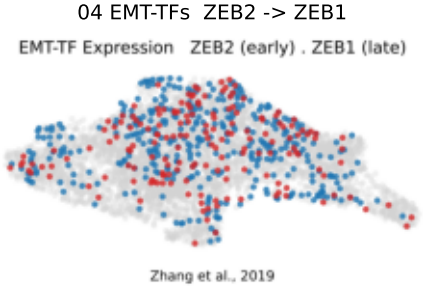
03 Somitogenesis (raw)



03 Neural Crest (raw)



04 EMT-TFs ZEB2 -> ZEB1



Dataset. Follicular lymphoma (Zhang et al., 2019), 10x; 4 tumour samples / 2 patients; 5,645 malignant B cells analysed.

Verdict. The embryonic pattern reproduces in its non-branching form: a single continuous gastrulation/somitogenesis -> neural-crest axis, rooted at the Gast/pSom-high pole, with EMT transcription factors ZEB2 (early) -> ZEB1 (late). No bifurcation, so no hub / Gate 3.

Parameter choices.

Sample threshold:	all 4 kept (min 680 >> floor; no gap cut - the 2x cliff is the 2-large/2-small split, not a thin-sample cliff)
Integration:	Harmony on sample, PCA(50), theta=10, n_pcs=15, spread=1.3 (theta from embedding + iLISI sweep)
Island removal:	radius graph 6x median-NN on UMAP; 0 cells removed (one connected component)
Geometry:	non-branching -> scFates curve (Nodes=20, lambda=0.3, mu=100); PPT tree tested = folds, no arm split
Root:	curve tip 0 = Gast/pSom-high, Crest-low (developmental origin; Crest is late)
EMT-TFs:	ZEB2 (early/root) <-> ZEB1 (late/terminus), detection-gated from a 10-factor panel